

REDUCTION OF SUPERELLIPTIC L -POLYNOMIALS

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ABSTRACT. Let $\mathbb{F}_q$ be a finite extension of $\mathbb{F}_p$. Let $\pi: C \rightarrow D$ be a finite morphism of proper, smooth, and geometrically connected curves over $\mathbb{F}_q$. Suppose π is generically cyclic of prime degree $r \neq p$. The numerators of the Hasse-Weil zeta functions of C and D are polynomials in $1 + T \cdot \mathbb{Z}[T]$. We give a congruence relating their reductions modulo r and apply it to numerators of zeta functions of superelliptic/hyperelliptic curves.

1. THEOREM STATEMENT

Let $\mathbb{F}_q$ be a finite extension of $\mathbb{F}_p$.

Let C be a proper, smooth, geometrically connected curve over $\mathbb{F}_q$.

Let $|C|$ be the set of closed points of C , that is, the set of $\text{Gal}(\bar{\mathbb{F}}_q/\mathbb{F}_q)$ -orbits of $C(\bar{\mathbb{F}}_q)$.

Let $Z(C, T) := \exp\left(\sum_{n=1}^{\infty} |C(\mathbb{F}_{q^n})| \frac{T^n}{n}\right) \in \mathbb{Q}(T)$ be the (Hasse-Weil) zeta function of C .

Let $L(C, T) := (1 - T)(1 - qT)Z(C, T) \in 1 + \mathbb{Z}[T]$ be the numerator of $Z(C, T)$.

Recall $Z(C, T) = \prod_{c \in |C|} (1 - T^{\deg(c)})^{-1}$ and $\deg(L(C, T)) = 2 \cdot \text{genus}(C)$.

Let $\text{Aut}_{\mathbb{F}_q}(C)$ be the group of automorphisms of C defined over $\mathbb{F}_q$.

Let $G \subseteq \text{Aut}_{\mathbb{F}_q}(C)$ be a finite subgroup of prime order $r \neq p$.

Let D be the quotient curve C/G and $\pi: C \rightarrow D$ be the quotient morphism.

Recall D is also proper, smooth, and geometrically connected over $\mathbb{F}_q$ and π is finite.

Let $Z \subset D$ be the branch locus of π and $|Z|$ be its set of closed points.

Theorem 1.

$$(1) \quad \frac{L(C, T)}{L(D, T)^r} (1 - T)^{r-1} (1 - qT)^{r-1} = \frac{Z(C, T)}{Z(D, T)^r} \equiv \prod_{z \in |Z|} (1 - T^{\deg(z)})^{r-1} \pmod{r}.$$

2. PROOF OF THEOREM

The identity in (1) uses the definition of $L(C, T), L(D, T)$.

For the congruence in (1), we start with the Euler product expansion

$$(2) \quad Z(C, T) = \prod_{c \in |C|} (1 - T^{\deg(c)})^{-1} = \prod_{d \in |D|} \prod_{c \in |C|: \pi(c)=d} (1 - T^{\deg(c)})^{-1}.$$

Let $|C_d| := \{c \in |C| : \pi(c) = d\}$ and $L(C_d, T) := \prod_{c \in |C_d|} (1 - T^{\deg(c)})$ for each $d \in |D|$.

Let $G_c \subseteq G$ be the stabilizer of $c \in |C|$.

Lemma 2. *Suppose $d \in |D| \setminus |Z|$ and $c \in |C_d|$.*

1. $\deg(c) = |G_c| \cdot \deg(d)$ and $||C_d|| = |G_c| \in \{1, r\}$.
2. $L(C_d, T) \in \{1 - T^{r \deg(d)}, (1 - T^{\deg(d)})^r\}$.
3. $L(C_d, T) \equiv (1 - T^{\deg(d)})^r \pmod{r}$.

Proof. Let $\mathbb{F}_q(c)$ be the splitting field of c , that is, $[\mathbb{F}_q(c) : \mathbb{F}_q] = \deg(c)$. Observe that G_c is the Galois group of the extension $\mathbb{F}_q(c)/\mathbb{F}_q(d)$ since π is unramified over d . Part (1) then follows from the orbit-stabilizer theorem for the action of G on $|C_d|$. It immediately implies (2) which in turn implies (3) since $(1 - T^{\deg(c)})^r \equiv (1 - T^{r \deg(c)}) \pmod{r}$. $\square$

Lemma 3. *Suppose $d \in |Z|$ and $c \in |C_d|$. Then $G_c = G$ and $L(C_d, T) = 1 - T^{\deg(d)}$.*

Proof. Observe π is totally ramified over d , hence $C_d = \{c\}$ and $\deg(c) = \deg(d)$. $\square$

Combining (2) and both lemmas gives

$$Z(C, T) \equiv \prod_{d \in |D|} (1 - T^{\deg(d)})^{-r} \prod_{z \in |Z|} (1 - T^{\deg(z)})^{r-1} \equiv Z(D, T)^r \cdot \prod_{z \in |Z|} (1 - T^{\deg(z)})^{r-1} \pmod{r}.$$

Divide both sides by $Z(D, T)^r$ to obtain the congruence in (1).

3. COROLLARIES

Proposition 4. $2 \cdot \text{genus}(C) = 2 + r \cdot (2 \cdot \text{genus}(D) - 2) + \sum_{z \in |Z|} \deg(z) \cdot (r - 1)$.

Proof. Take degrees of both sides in (1) (or apply the Riemann-Hurwitz formula). $\square$

Proposition 5. *Suppose that $\text{genus}(D) = 0$. Then*

$$L(C, T) \equiv (1 - T)^{1-r} (1 - qT)^{1-r} \prod_{z \in |Z|} (1 - T^{\deg(z)})^{r-1} \pmod{r}.$$

Proof. Observe that $L(D, T) = 1$ and that $q \equiv 1 \pmod{2}$, and rearrange (1). $\square$

4. EXAMPLE

Let $f \in \mathbb{F}_q[x]$ be square-free of odd degree $2g + 1$.

Let $f = \prod_{i=1}^m \pi_i \in \mathbb{F}_q[x]$ be a prime factorization.

Let U the affine curve $y^2 = f(x)$ and $C := U \sqcup \{\infty\}$ be its smooth completion.

Let $G = \boldsymbol{\mu}_2 = \{\pm 1\} \subseteq \mathbb{F}_q^\times$ act on C by $-(x, y) := (x, -y)$ and $-\infty := \infty$.

Observe $D \cong \mathbb{P}^1$ is the x -line and $\pi: C \rightarrow D$ is given by $(x, y) \mapsto [x, 1]$ and $\infty \mapsto \infty$.

Deduce $\text{genus}(D) = 0$ and $\text{genus}(C) = g$. Moreover,

$$L(C, T) \equiv (1 - T)^{-1} \prod_{i=1}^m (1 - T^{\deg(\pi_i)}) \pmod{2}.$$

In particular, if $m = 3$ and $(\deg(\pi_1), \deg(\pi_2), \deg(\pi_3)) = (1, g, g)$, then

$$L(C, T) \equiv (1 - T^g)^2 \equiv 1 - T^{2g} \pmod{2}.$$